

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458717.8051 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.021
Transit depth (Rp/Rs)^2	1.46 +/- 0.51 [%]
Orbital Inclination (inc)	84.38 +/- 1.21
Airmass coefficient 1 (a1)	1.246 +/- 0.013
Airmass coefficient 2 (a2)	-0.188 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	None
Optimal Aperture	4.54
Optimal Annulus	17.38
Transit Duration (day)	0.064 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.121 ± 0.021 vs 0.1326 (deviation 0.53σ)
Duration fit vs published	0.064 d vs 0.0737 d (deviation 0.47σ)
Mid-time O–C	14.8 min
Depth SNR	5.6
Residual scatter	1.01 %

Light curve

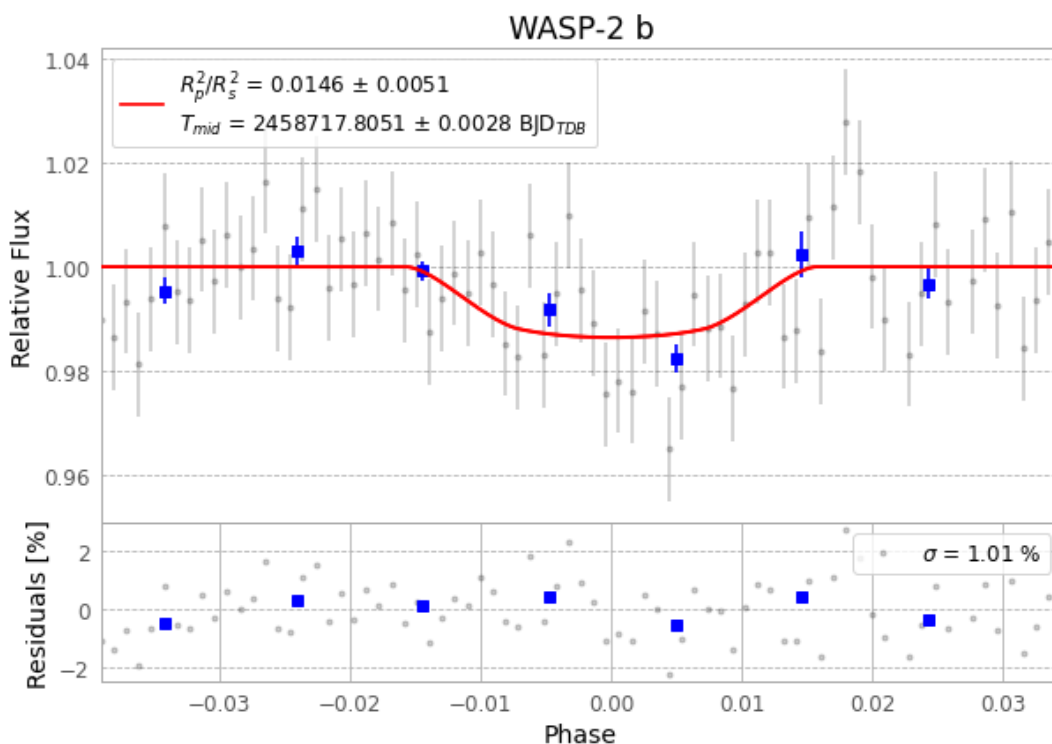


Figure 1 — FinalLightCurve_WASP-2 b_2019-08-21.png (SHA-256 04083e64852ad591...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [171, 193], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ef65fb704547b0cf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-2190822050912.FITS → WASP-2190822085712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-190822.log (a729af6b19ac...), FinalLightCurve_WASP-2 b_2019-08-21.png (04083e64852a...), FinalLightCurve_WASP-2 b_2019-08-21.csv (ce1e3a5d98b0...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 20:40Z.